# Exoplanet Detection and Habitability Analysis

Team Name - Team Name

***Abstract*—This report details the photometric analysis of three distant star systems (STAC-1, STAC-2, and STAC-3) to identify potential habitable exoplanets for a new Alliance base. By calculating transit depths, orbital periods, planetary radii, and orbital distances, we confirm the presence of three distinct exoplanets. Our habitability analysis concludes that STAC-3b is the best candidate to establish a surface outpost.**

## I. Detection of Exoplanets

A periodic decrease in a star's relative brightness indicates that an orbiting celestial body passed between the star and the observer physically obstructing a fraction of the light from the star. We analyzed the periodic brightness data of three different stars. The transit depth ($\delta$) is calculated as:

$$\delta = \frac{F_{\text{normal}} - F_{\text{transit}}}{F_{\text{normal}}} \tag{1}$$

*1) STAC-1 system:* The brightness drops from a constant 1.0000 to a minimum of 0.9970 on days 3, 8, 12, 16, and 20. This is clearly a repeating transit. The total elapsed time for 4 orbits in 17 days giving an average orbital period ($P$) of

$$P = \frac{17}{4} = 4.25 \text{ days}$$

TABLE I
STAC-1 — Brightness Observations

| Time (days) | Rel. brightness | Time (days) | Rel. brightness | Time (days) | Rel. brightness |
|---|---|---|---|---|---|
| 0 | 1.0000 | 7 | 1.0000 | 14 | 1.0000 |
| 1 | 1.0000 | 8 | 0.9970 | 15 | 1.0000 |
| 2 | 1.0000 | 9 | 1.0000 | 16 | 0.9970 |
| 3 | 0.9970 | 10 | 1.0000 | 17 | 1.0000 |
| 4 | 1.0000 | 11 | 1.0000 | 18 | 1.0000 |
| 5 | 1.0000 | 12 | 0.9970 | 19 | 1.0000 |
| 6 | 1.0000 | 13 | 1.0000 | 20 | 0.9970 |

$$\delta = \frac{1.0000 - 0.9970}{1.0000}$$

$$\delta = 0.0030$$

The maximum transit depth is $\delta = 0.0030$

*2) STAC-2 system:* We observe the brightness drops on days 20, 100 and 180 which is also clearly a repeating transit. The total elapsed time for 2 orbits in 160 days giving an average orbital period ($P$) of

$$P = \frac{160}{2} = 80.0 \text{ days}$$

TABLE II
STAC-2 — Brightness Observations

| Time (days) | Rel. brightness | Time (days) | Rel. brightness |
|---|---|---|---|
| 0 | 1.0000 | 120 | 1.0000 |
| 20 | 0.9990 | 140 | 1.0000 |
| 40 | 1.0000 | 160 | 1.0000 |
| 60 | 1.0000 | 180 | 0.9990 |
| 80 | 1.0000 | 200 | 1.0000 |
| 100 | 0.9990 | | |

$$\delta = \frac{1.0000 - 0.9990}{1.0000}$$

$$\delta = 0.0010$$

The maximum transit depth is $\delta = 0.0010$

*3) STAC-3 System:* The brightness drops from 1.0000 to a minimum of 0.9990 on day 20 and day 170 respectively. So, we take them as near-centred transits separated by 150 days. Testing for n = 1 to 6, only n = 4 reproduces both dips at days 57.5, 95 and 132.5 which justifies irregular Rel. brightness at days 60, 100 and 140 and flat readings elsewhere. This time interval yields an orbital period (P) of

$$\frac{150}{4} = 37.5 \text{ days}$$

TABLE III
STAC-3 — Brightness Observations

| Time (days) | Rel. brightness | Time (days) | Rel. brightness | Time (days) | Rel. brightness |
|---|---|---|---|---|---|
| 0 | 1.0000 | 70 | 1.0000 | 140 | 0.9993 |
| 10 | 1.0000 | 80 | 1.0000 | 150 | 1.0000 |
| 20 | 0.9990 | 90 | 1.0000 | 160 | 1.0000 |
| 30 | 1.0000 | 100 | 0.9991 | 170 | 0.9990 |
| 40 | 1.0000 | 110 | 1.0000 | 180 | 1.0000 |
| 50 | 1.0000 | 120 | 1.0000 | 190 | 1.0000 |
| 60 | 0.9992 | 130 | 1.0000 | 200 | 1.0000 |

$$\delta = \frac{1.0000 - 0.9990}{1.0000}$$

$$\delta = 0.0010$$

The maximum transit depth is $\delta = 0.0010$

## II. Determination of Planet Size

Transit depth is equal to the ratio of the planet's cross-sectional area to the star's cross-sectional area.

$$\delta = \left(\frac{R_{\text{p}}}{R_{\text{s}}}\right)^2$$

Rearranging this, we get:

$$R_p = R_s \times \sqrt{\delta} \tag{2}$$

Calculations use the standard Solar radius ($R_\odot = 696,000$ km) and Earth radius ($R_\oplus = 6,371$ km).

*1) STAC-1b:* Given $R_s = 1.00R_\odot$ (696,000 km) and $\delta = 0.0030$:

$$R_p = 696,000 \times \sqrt{0.0030} = 38,121 \text{ km}$$

Dividing by Earth's radius yields $5.98R_\oplus$. This body is classified as a **Neptune-sized** gas giant.

*2) STAC-2b:* Given $R_s = 0.80R_\odot$ (556,800 km) and $\delta = 0.0010$:

$$R_p = 556,800 \times \sqrt{0.0010} = 17,608 \text{ km}$$

This is equivalent to $2.76R_\oplus$. It is classified on the boundary of a massive **Super-Earth** or a Mini-Neptune.

*3) STAC-3b:* Given $R_s = 0.50R_\odot$ (348,000 km) and $\delta = 0.0010$:

$$R_p = 348,000 \times \sqrt{0.0010} = 11,005 \text{ km}$$

This equates to $1.73R_\oplus$, classifying it firmly as a rocky **Super-Earth**.

## III. Planetary Orbits

Orbital distance ($a$) is calculated using the simplified form of Kepler's Third Law.

$$P^2 = \frac{a^3}{M}$$

On simplification, we get:

$$a = \sqrt[3]{P^2 \times M} \tag{3}$$

*1) STAC-1b ($M = 1.00M_\odot$):* Period = 0.0116 years.

$$\text{Distance} = \sqrt[3]{0.0116^2 \times 1.00} = 0.051 \text{ AU}$$

This equates to 7.63 million km, placing the planet roughly 20 times closer to its star than Earth is to the Sun.

*2) STAC-2b ($M = 0.80M_\odot$):* Period = 0.219 years.

$$\text{Distance} = \sqrt[3]{0.219^2 \times 0.80} = 0.337 \text{ AU}$$

This is 50.4 million km, approximately one-third of the Earth-Sun distance.

*3) STAC-3b ($M = 0.50M_\odot$):* Period = 0.103 years.

$$\text{Distance} = \sqrt[3]{0.103^2 \times 0.50} = 0.174 \text{ AU}$$

At 26.0 million km, it orbits at roughly 17% of Earth's distance from the Sun.

## IV. Habitability Analysis and Conclusion

The theoretical Habitable Zone (HZ) distance for each host star is approximated by:

$$d_{\text{HZ}} \approx \sqrt{L/L_\odot} \text{ AU}$$

The calculated distances for the three planets are:

*1) STAC-1b:* L = $1.00L_\odot$, $d_{\text{HZ}} \approx \sqrt{1.00/1.00}$ AU = 1.00 AU

*2) STAC-2b:* L = $0.40L_\odot$, $d_{\text{HZ}} \approx \sqrt{0.40/1.00}$ AU = 0.63 AU

*3) STAC-1b:* L = $0.08L_\odot$, $d_{\text{HZ}} \approx \sqrt{0.08/1.00}$ AU = 0.28 AU

Table IV summarises all such parametric measurements of planets STAC-1b, STAC-2b and STAC-3b.

Based on our analysis of table IV, **STAC-3b** is the only favorable planet to establish a new base. The three primary evidences which support this conclusion are:

1) **Favorable Composition:** $1.73R_\oplus$ indicates that STAC-3b is a Super-Earth. Planetary astrophysics indicates that bodies of this radius are largely composed of rock and metal meaning it possesses the solid surface required for a base. On the other hand, STAC-1b and STAC-2b are gaseous giants.
2) **Habitable Zone:** STAC-3b orbits at $0.174$ AU, which is comparatively closer to its star's theoretical habitable boundary of $0.28$ AU than STAC-1b $(0.051)$ and STAC-2b $(0.337)$. The table also suggests STAC-1b is severely irradiated.
3) **Stability:** The host of STAC-3 is an M-dwarf star. These low-mass stars offer incredible long-term stability and longevity, providing a safe, enduring environment.

TABLE IV
SUMMARY OF PLANETARY PHYSICAL AND ORBITAL PROPERTIES

| **System** | **Period (days)** | **Transit Depth ($\delta$)** | **Radius ($R_\oplus$)** | **Classification** | **Orbit (AU)** | **Star HZ (AU)** |
|---|---|---|---|---|---|---|
| STAC-1b | 4.25 | 0.0030 | 5.98 | Neptune-sized | 0.051 | 1.00 |
| STAC-2b | 80.0 | 0.0010 | 2.76 | Super-Earth / Mini-Neptune | 0.337 | 0.63 |
| STAC-3b | 37.5 | 0.0010 | 1.73 | Super-Earth | 0.174 | 0.28 |